

HyLaT: Efficient Multi-Agent Communication via Hybrid Latent-Text Protocol

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Abstract

Communication protocol design is a central challenge in large language model-based multi-agent systems. Existing single-channel approaches face an inherent communication trilemma: text-based methods are interpretable but verbose, while latent-space methods are efficient but opaque and limited to unidirectional workflows. Inspired by multi-channel communication theory, we propose **HyLaT**, a hybrid latent-text communication protocol that transmits elaborate cognitive signals through a latent channel for efficiency, while expressing concise critical signals in natural language to preserve interpretability and precision. We introduce a two-stage training framework combining single-agent hybrid generation learning and multi-agent interactive co-training, enabling agents to generate and interpret hybrid messages across multiple rounds of interaction. Experiments demonstrate that HyLaT reduces communication overhead significantly while maintaining competitive task performance, with strong generalization and robustness across diverse settings¹.

1 Introduction

Large Language Models (LLM)-based multi-agent systems have shown strong performance in collaborative problem solving and social simulations (Guo et al., 2024; Mou et al., 2024a; Qian et al., 2024a). In these systems, communication serves as the backbone of coordination, enabling agents to exchange information, align on shared goals, and collectively reason across extended interactions. As systems scale to more agents and longer horizons, the design of inter-agent communication protocols becomes a central challenge that directly

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¹Code is available at <https://github.com/xymou/hylat>.

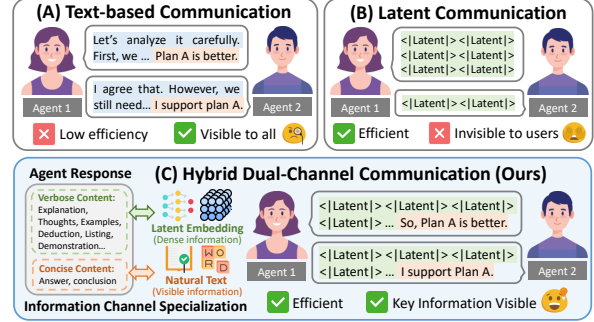


Figure 1: **Comparison among different multi-agent communication paradigms.** (A) Text-based communication: fully readable but with a heavy efficiency bottleneck. (B) Latent communication: highly efficient but opaque to users. (C) Hybrid latent-text communication (ours): balances efficiency and explainability through a dual-channel design.

shapes system capability and scalability (Marro et al., 2024; Zhang et al., 2024; Chen et al., 2025).

Existing communication paradigms can be broadly characterized as **single-channel**, either text or latent space. As illustrated in Figure 1A, text-based methods (Du et al., 2023; Park et al., 2023; Zhang et al., 2025) rely entirely on natural language, which, while interpretable, leads to verbose messages with high token overhead. Recent latent-space communication methods (Zou et al., 2025; Du et al., 2025; Fu et al., 2025; Wu et al., 2025) instead exchange internal representations (Figure 1B), achieving dense and efficient transfer but at the cost of interpretability. Latent signals are opaque to external observers, limiting applicability to scenarios where intermediate transparency is essential, such as social simulation. Moreover, existing latent-space methods are primarily designed for single-shot or one-directional exchanges within a fixed workflow. This leaves multi-round interactive communication among multiple agents in more realistic settings largely unsupported.

The tension between interpretability and efficiency reflects a deeper structural limitation known

as the *agent communication trilemma* (Marro et al., 2024): no single-channel protocols can simultaneously optimize for efficiency, portability, and versatility. This suggests that the communication channel itself must be reconsidered. Inspired by multi-channel communication theory (Shannon, 1948; Monge and Contractor, 2003) that different types of information are best transmitted through varied channels suited to their functional roles, we argue that the heterogeneous nature of agent messages calls for an analogous hybrid communication design. Specifically, agent messages often contain two functionally distinct types of information: (1) **elaborate cognitive signals**, such as explanations, intermediate reasoning, and contextual elaboration, that are well-suited to compact latent-space representations; and (2) **concise but critical signals**, such as answers, commitments, or final decisions, that must remain interpretable to both agents and human observers.

To this end, we propose **HyLaT**, a **Hybrid Latent-Text** communication protocol that routes each type of information through a dedicated channel (Figure 1C). Dense intermediate information flows through a latent channel for efficient machine-to-machine transfer, while concise critical signals are expressed in natural language to preserve interpretability. This dual-channel design addresses all axes of the trilemma: the latent channel provides efficiency, the text channel enables interoperability with agents using other protocols, and the combination provides versatility by supporting multi-round, multi-agent interaction, a setting that prior latent-space methods largely cannot handle.

Realizing this design poses key training challenges that agents need to learn generating hybrid outputs and also interpreting latent vectors from peers during multi-turn interactions. We address this with a two-stage training framework. The first stage, *single-agent hybrid generation learning*, equips the model to generate hybrid responses from text-only inputs, using a cross-channel alignment loss to transfer the semantic content of textual elaborations into the latent space. The second stage, *multi-agent interactive co-learning*, enables agents to jointly learn to interpret and respond across both channels. We evaluate HyLaT on various tasks, demonstrating that it significantly reduces communication overhead while maintaining strong task performance and preserving interpretability. Our contributions are summarized as follows:

- We identify channel multiplicity as a structural resolution to the agent communication trilemma and introduce HyLaT, a hybrid latent-text communication protocol that enables agents to exchange information through complementary latent and textual channels.
- We propose a two-stage training approach combining single-agent hybrid learning and multi-agent interactive co-learning, enabling agents to generate and interpret both latent and textual messages in multi-round interactions.
- Experiments demonstrate that HyLaT effectively balances communication efficiency and information transparency, enabling scalable and interpretable multi-agent collaboration.

2 Methodology

2.1 Overview

This section details the proposed HyLaT framework, designed to facilitate efficient and transparent communication among agents. As illustrated in Figure 2, we first define the **dual-channel agent communication protocol** in Section 2.2. Furthermore, we design a two-stage training framework to empower existing models with dual-channel communication capabilities. **Stage 1: Single-Agent Hybrid Generation Learning** (Section 2.3) trains each agent to generate hybrid outputs consisting of both text and latent representations; and **Stage 2: Multi-agent Interactive Co-Training** (Section 2.4) enables the shared backbone of multiple agents to jointly learn to interpret and respond to dual-channel information generated by others during multi-turn interactions.

2.2 Hybrid Communication Protocol

Interactive Setup As illustrated in the top part of Figure 2, we consider a multi-agent system of N agents powered by a shared backbone model M , communicating over T interaction rounds. At round $t = 1$, every agent receives a text context. At each subsequent round $t > 1$, agent i receives its accumulated context together with the responses from all other agents in the previous round:

$$x_i^{(t)} = \left[\mathbf{x}_i^{(t-1)}, \oplus_{j \neq i} o_j^{(t-1)} \right], \quad t > 1 \quad (1)$$

where $\mathbf{x}_i^{(t-1)}$ is agent i 's accumulated context up to round $t - 1$, $o_j^{(t-1)}$ denotes the response from agent j at round $t - 1$, and $\oplus$ denotes concatenation. Agent i then generates a response $o_i^{(t)} = M(x_i^{(t)})$.

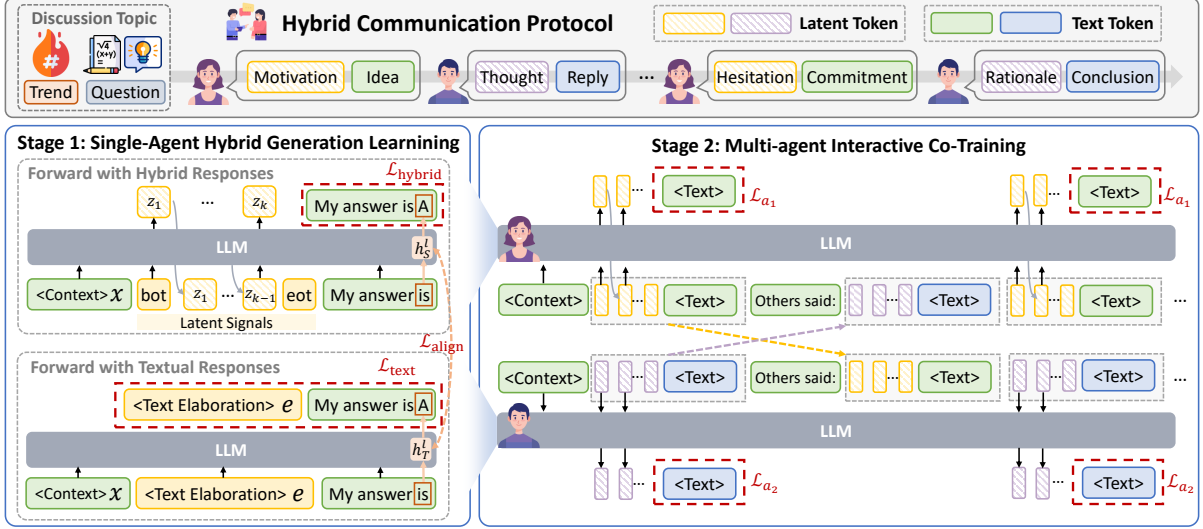


Figure 2: Overview of the proposed HyLaT framework. Agents exchange elaborate intermediate signals through a latent channel and concise commitments through a text channel. The model is trained to generate hybrid content in Stage 1 and to support multi-round hybrid communication in Stage 2.

Dual-Channel Response Format In HyLaT, the output of agent i at round t consists of two parts from distinct channels: a latent component $\mathbf{Z}_i^{(t)} = (z_1, \dots, z_k)$ carrying elaborate cognitive signals, and a text component $\mathbf{Y}_i^{(t)} = (y_1, \dots, y_m)$ carrying concise, interpretable conclusions:

$$o_i^{(t)} = \underbrace{[\langle \text{bot} \rangle z_1, \dots, z_k \langle \text{eot} \rangle]}_{\text{latent component}} \parallel \underbrace{[y_1, \dots, y_m]}_{\text{text component}}, \quad (2)$$

where the **latent component** consists of k continuous vectors produced by autoregressively propagating the model’s last hidden state, delimited by two learnable special tokens $\langle \text{bot} \rangle$ and $\langle \text{eot} \rangle$. The **text component** is a standard natural-language sequence generated after $\langle \text{eot} \rangle$.

Channel Specialization To balance communication efficiency with transparency, we retain the summary part of agent responses—such as final answers and commitments—in the textual channel. Conversely, verbose information, including explanations, elaboration, and reasoning processes, is compressed into the latent component. This channel specialization capability is induced through the structure of training data described in Section 2.3.

2.3 Stage 1: Single-Agent Hybrid Generation Learning

The goal of Stage 1 is to train the model to produce well-formed hybrid outputs from text-only input, preparing it for multi-agent interaction in Stage 2. To achieve this, we construct single- and multi-turn user-assistant dialogues in single-agent scenarios

as supervision. Each turn pairs a text input x with a target response including both a textual elaboration e and a critical output y , corresponding to the two information types defined earlier.

As shown in Figure 2 (bottom-left), each training step involves two forward passes over the input:

Forward with Hybrid Responses Since no ground-truth latent supervision exists, teacher forcing is inapplicable to the latent component. The model instead autoregressively generates k continuous latent vectors ($\mathbf{Z} = \{z_i\}_{i=1}^k$) starting from $\langle \text{bot} \rangle$, after which $\langle \text{eot} \rangle$ and target y are appended. The language modeling loss is computed over y :

$$\mathcal{L}_{\text{hybrid}} = -\frac{1}{|y|} \sum_i \log P(y_i | y_{1:i-1}, x, \mathbf{Z}), \quad (3)$$

$\mathcal{L}_{\text{hybrid}}$ guides the model to derive y based on its self-generated intermediate latent information $\mathbf{Z}$.

Forward with Textual Responses To provide meaningful supervision for $\mathbf{Z}$, we follow Shen et al. (2025) and introduce a pure-text branch. In parallel with the hybrid forward pass, the model is trained to generate $r = [e, y]$ conditioned on x :

$$\mathcal{L}_{\text{text}} = -\frac{1}{|r|} \sum_i \log P(r_i | r_{1:i-1}, x), \quad (4)$$

$\mathcal{L}_{\text{text}}$ is optimized under teacher forcing to preserve the model’s ability to produce explicit elaborations.

Cross-Channel Alignment Based on the parallel forward passes, we transfer the information encoded in e into the latent space by aligning the

hidden activations of both branches at a shared target position. This position is defined as the token immediately preceding the final answer (e.g., the token “is” in “My answer is A”), where all preceding context about the elaboration must be compressed before predicting the answer. This alignment is applied across all L transformer layers:

$$\mathcal{L}_{\text{align}} = \frac{1}{L} \sum_{l=1}^L \left\| \text{sg} \left[\mathbf{h}_T^l \right] - \mathbf{h}_S^l \right\|_2, \quad (5)$$

where $\mathbf{h}_T^l$ and $\mathbf{h}_S^l$ denote the hidden activations at the target position in the l -th layer for the textual and the hybrid forward pass, respectively. $\text{sg}(\cdot)$ is the stop-gradient operation.

Overall, the training loss for Stage 1 is:

$$\mathcal{L}_{\text{stage1}} = \alpha \mathcal{L}_{\text{hybrid}} + \beta \mathcal{L}_{\text{text}} + \gamma \mathcal{L}_{\text{align}}, \quad (6)$$

where α , β and γ are hyperparameters. This training process can be directly extended to multi-turn dialogues by generating latent-channel responses turn-by-turn during the hybrid forward pass.

2.4 Stage 2: Multi-Agent Interactive Co-Training

Stage 2 trains agents to communicate under the HyLaT protocol, exposing each agent to latent vectors produced by its peers. The training data consists of multi-turn, multi-agent dialogues, including the topic context for the initial round $x^{(1)}$ and the textual responses $\{y_i^{(t)}\}_{i=1}^N$ from N agents at each turn t . Similar to Stage 1, each response $y_i^{(t)}$ is augmented with its corresponding elaboration $e_i^{(t)}$.

Following the protocol formulated in Section 2.2, we construct the input $x_i^{(t)}$ for each agent at turn t by integrating prior context and responses from other agents (Equation 1). Regarding the outputs, consistent with Stage 1, the textual forward pass utilizes the labeled $[e_i^{(t)}, y_i^{(t)}]$ under teacher-forcing training, while the hybrid forward pass generates latent representations $Z_i^{(t)}$ turn-by-turn.

Based on the constructed inputs and outputs, we calculate the loss $\mathcal{L}_{\text{stage1}}$ for each agent, then aggregate losses across all N agents:

$$\mathcal{L}_{\text{stage2}} = \frac{1}{N} \sum_{i=1}^N \left(\alpha \mathcal{L}_{\text{hybrid}}^{(i)} + \beta \mathcal{L}_{\text{text}}^{(i)} + \gamma \mathcal{L}_{\text{align}}^{(i)} \right), \quad (7)$$

where $\mathcal{L}_{\text{hybrid}}^{(i)}$, $\mathcal{L}_{\text{text}}^{(i)}$, and $\mathcal{L}_{\text{align}}^{(i)}$ are respectively defined in Equations 3–5. Through this joint training,

agents are exposed to peers’ latent vectors as part of their input context and learn to interpret them while generating a proper hybrid response.

3 Experimental Setup

3.1 Training Data Construction

Stage 1 To construct data required in Section 2.3, we collect datasets with detailed explanations or reasoning chains as supervision for the latent component: CommonsenseQA (Talmor et al., 2019), SocialIQA (Sap et al., 2019), WorldTree (Jansen et al., 2018), PubMedQA (Jin et al., 2019), and StrategyQA (Geva et al., 2021). All selected datasets contain detailed reasoning processes paired with concise final answers. This data-driven approach allows channel specialization to emerge naturally from the structure of the training examples, without requiring explicit routing annotations.

Stage 2 We construct two types of multi-agent interaction data. (1) *Refinement*: using Llama-3.2-1B-Instruct and GPT-5 (Singh et al., 2025) to conduct simulated multi-agent debate on Stage 1 datasets, producing multi-round hybrid-output trajectories that converge to a final answer, where the answers of the first round may be incorrect. (2) *Decomposition*: Multi-hop questions sampled from HotpotQA (Yang et al., 2018), 2WikiMultiHopQA (Ho et al., 2020), and GSM8K-AugNL (Shen et al., 2025) are decomposed into parallel sub-questions assigned across agents, who collaborate to reach a final answer. Dataset statistics and construction details are provided in Appendix A.1.

3.2 Implementation Details

We use Llama-3.2-1B-Instruct (Grattafiori et al., 2024) as the backbone for main experiments, while experiments on other backbones are provided in Appendix B. Parameter-efficient fine-tuning is applied via LoRA (Hu et al., 2022) with rank $r = 128$ and $\alpha = 32$, applied to all attention projection matrices. The number of latent vectors per turn is set to $k = 6$. Both stages are trained on 2-turn dialogues, with Stage 2 involving $N=2$ agents. All training experiments are conducted on 8 NVIDIA H200 GPUs. For evaluation, each task is run independently on a single H200 GPU. More details can be found in Appendix A.3.

3.3 Evaluation

Tasks Following Du et al. (2023), we adopt multi-agent debate (MAD) as the primary task for

Method	In-Domain										Out-of-Domain						Inference Efficiency	
	Commonsense		StrategyQA		SocialIQA		WorldTree		PubMedQA		MedQA		ARC-E		ARC-C		# token	time
	Avg.	Maj.	Avg.	Maj.	Avg.	Maj.	Avg.	Maj.	Avg.	Maj.	Avg.	Maj.	Avg.	Maj.	Avg.	Maj.		
Training-Free Single-Agent Baseline																		
Single-Agent	43.67	43.67	52.00	52.00	56.00	56.00	59.33	59.33	55.67	55.67	38.33	38.33	30.67	30.67	34.00	34.00	113.59	1.25
Training-Free Text-Based Communication Methods																		
NL	49.56	53.33	54.44	54.67	58.22	61.00	69.33	73.00	54.56	59.00	42.67	42.67	35.33	37.67	31.78	32.00	1247.50	13.15
AutoForm	47.78	50.33	54.56	55.00	59.89	63.00	67.11	68.67	56.11	57.00	44.11	46.00	35.89	35.33	33.56	34.00	1717.60	18.18
EcoLang	48.00	50.67	49.44	50.33	55.78	61.33	68.11	71.33	54.67	57.33	41.89	42.67	37.33	36.67	34.89	36.00	960.26	10.04
SDE	47.22	49.00	52.67	52.67	58.11	60.67	64.56	68.00	49.67	53.00	40.78	41.67	34.78	36.67	32.89	32.33	1316.20	14.48
Training-Free Latent-Space Communication Methods																		
Cipher	50.22	52.00	52.78	54.67	57.89	60.33	69.22	71.33	55.44	57.67	42.56	43.33	35.67	36.00	34.56	35.33	1137.89	11.58
LatentMAS-V	26.11	26.00	49.67	50.00	42.78	43.00	45.00	46.33	54.22	55.00	32.67	33.00	27.67	27.89	27.00	28.33	324.43	3.61
Training-Based Communication Methods																		
TextFullIT	64.44	66.33	64.11	65.00	76.66	77.63	70.56	73.33	69.67	71.00	44.55	44.67	42.33	42.67	34.89	34.67	505.03	5.47
LatentFullIT	56.00	56.33	63.11	63.33	80.78	80.67	68.56	68.33	66.11	66.67	42.11	42.33	38.89	38.67	33.78	34.00	57.00	0.70
HyLaT	63.78	64.33	64.67	65.33	82.56	83.00	69.22	69.00	67.44	67.33	42.11	42.33	39.00	39.33	33.89	34.00	72.01	1.47

Table 1: Experimental results of different methods. We report average accuracy (Avg.) and majority-vote accuracy (Maj.) across agents, along with communication efficiency measured by average token count (# token) and wall-clock time (time, in seconds). **Bold** denotes the best result and underline denotes the second best.

evaluation. We report results on both in-domain datasets encountered during training, i.e., CommonsenseQA, StrategyQA, SocialIQA, WorldTree, and PubMedQA, as well as several out-of-domain datasets, specifically MedQA (Jin et al., 2021), ARC-Easy and ARC-Challenge (Clark et al., 2018). We follow Tang et al. (2025) to use the first 300 questions of each dataset to build MAD tasks. We conduct evaluations on these tasks with 3 agents communicating over 2 turns.

Metrics We assess both task performance and communication efficiency. For **performance**, we report the average per-agent accuracy (Avg.) across all agents and the majority-vote accuracy (Maj.) aggregated over all agents. For **efficiency**, we report the average number of tokens generated per question across all communication rounds (# token), and the average running time per question.

3.4 Compared Methods

We compare HyLaT against various methods designed to optimize communication between agents. The “**Single-Agent**” baseline serves as a lower bound with no inter-agent communication. Multi-agent baselines are further categorized into: (1) text-based methods, including:

- **NL** uses unoptimised Natural-Language messages as the communication channel;
- **AutoForm** (Chen et al., 2024) directly prompts agents to communicate in structured language, such as JSON and code;
- **EcoLang** (Mou et al., 2025) evolves efficient communication rules via a selection procedure and prepends them to agent instructions;

- **SDE** (Tang et al., 2025), where agents transmit textual tokens alongside the model’s internal state-difference trajectory recorded in generation; and (2) latent-space methods, including:

- **Cipher** (Pham et al., 2023) replaces textual messages with semantic embedding vectors, computed as the belief-weighted average over the output vocabulary distribution.

- **LatentMAS-V²** (Zou et al., 2025) has each agent autoregressively generate hidden-state vectors from the final layer and pass its full key–value cache as a latent memory to the next agent.

Since the aforementioned methods do not involve additional training, we introduce additional baselines, **TextFullT** and **LatentFullT**, to ensure a fair comparison. **TextFullT** is trained on the same data as HyLaT but restricts all communication, including elaborations and conclusions to a single textual channel. **LatentFullT** is trained on the same data as HyLaT, but intermediate rounds communicate purely through latent vectors, while only the final round retains a latent vector and a text output for supervision and evaluation.

4 Experimental Results

4.1 Main Results

Table 1 reports the overall performance of various methods, with several key findings highlighted:

I. Training-free methods fail to balance efficiency and performance. Among training-free

²Since the original LatentMAS does not distinguish between dialogue roles and does not support multi-round multi-agent communication, we adapt its implementation to the multi-round setting. This modification may not reflect its intended use and could underestimate its performance.

Method	Avg.	Maj.	# token	time	format err.(%)
TextFullT	58.40	59.41	505.03	5.47	0.00
LatentFullT	56.17	56.29	57.00	0.70	0.00
Ours	57.83	58.08	72.01	1.47	0.00
<i>Variants based on Single-Channel Communication</i>					
<i>w/ pure text</i>	56.93	57.00	639.18	13.18	3.28
<i>w/ pure latent</i>	50.28	50.33	56.09	1.16	6.12
<i>Variants with Single-Stage Training</i>					
<i>w/o Stage 1</i>	38.43	38.58	102.59	1.38	9.13
<i>w/o Stage 2</i>	43.64	43.96	136.10	2.87	22.27

Table 2: Ablation study of the HyLaT framework. Reported results are averaged over all datasets.

approaches: (1) The single-agent baseline achieves the highest efficiency but suffers from poor performance. (2) Text-based multi-agent methods yield strong results, yet offer limited room for efficiency optimization—even the most efficient one, EcoLang, requires $9\times$ the duration of the single-agent baseline. (3) Latent-space methods are generally more efficient, particularly LatentMAS-V, but at the cost of a severe performance drop. These observations suggest that in a zero-shot setting, the fixed backbone model significantly constrains agent behavior, highlighting the **necessity of additional training to co-optimize efficiency and performance**.

II. HyLaT strikes a balance between efficiency and performance. (1) **Unparalleled efficiency.** HyLaT operates at execution times close to the single-agent baseline while using only half the tokens. Specifically, HyLaT improves token and time efficiency by $10.6\times/6.8\times$ and $3.6\times/2.5\times$ over the most efficient text-based and latent-space baselines, respectively. (2) **Task performance.** Unlike LatentMAS-V, HyLaT preserves strong task performance, achieving top-2 or comparative performance on both in-domain and out-of-domain datasets. Notably, on SocialIQA, HyLaT clearly outperforms all baselines by utilizing the latent channel to supplement information that text alone cannot fully convey. (3) **Advantage over Text-FullT and LatentFullT.** Although HyLaT slightly trails TextFullT on tasks requiring exhaustive structured reasoning, such as WorldTree and ARC-E, its overall performance is highly comparable, while incurring substantially lower time and token costs. Meanwhile, LatentFullT only slightly improves computational efficiency, but its performance is inferior to that of HyLaT, suggesting that the intermediate text channel serves as a semantic anchor that maintains contextual coherence across rounds. In summary, HyLaT effectively enhances multi-agent

communication efficiency, **maintaining task capabilities of other multi-agent frameworks while operating at or below single-agent costs.**

4.2 Ablation Study

Furthermore, we compare several variants of HyLaT in Table 2 to validate the effectiveness of the proposed dual-channel communication framework.

Effect of the Hybrid Channel Design We first ablate the channel composition by comparing two degenerate variants: (1) *w/ pure text* uses our trained model, rather than TextFullT, to generate natural language throughout all rounds, resulting in significantly higher token usage and inference time, while yielding only marginal accuracy gains. This confirms that **the latent channel is the primary driver of HyLaT’s efficiency advantage**. (2) *w/ pure latent* restricts intermediate rounds to latent-only outputs, reserving hybrid generation for the final round only. Despite achieving the lowest token count, this variant suffers a notable drop in accuracy and a higher format error rate, suggesting that **textual signals are crucial for agents to maintain coherent comprehension across multiple turns**. In summary, the advantages of the two channels are clearly complementary. Thus, the dual-channel design of HyLaT represents the optimal choice for balancing efficiency and performance.

Effect of Two-stage Training The variant *w/o Stage 1* skips the single-agent warm-up and directly trains multi-agent interaction from scratch. Without first learning to generate stable hybrid outputs at the individual level, the latent representations produced early in training are noisy and unstable, and these errors propagate across rounds and agents, making optimization substantially more difficult and leading to negative results. The variant *w/o Stage 2* applies the Stage 1 checkpoint directly to multi-agent communication without co-learning. Since the model has never been exposed to peers’ hybrid outputs during training, it frequently fails to process latent vectors from other agents at test time, resulting in a high format error rate (22.27%) and a substantial drop in task performance. This confirms that Stage 2 co-learning is essential for enabling robust multi-round hybrid communication.

4.3 Compatibility and Robustness of the Hybrid Protocol

Communication Protocol Compatibility A practical advantage of HyLaT’s hybrid design is

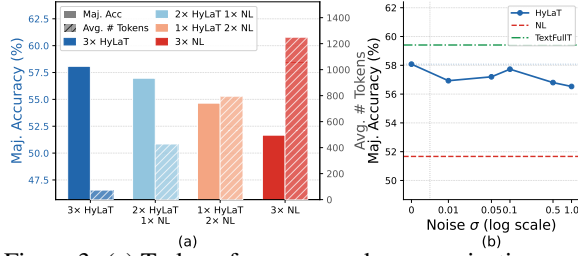


Figure 3: (a) Task performance and communication cost under heterogeneous agent settings; (b) Majority voting accuracy of HyLaT under increasing Gaussian noise applied to latent vectors at inference time.

that the preserved text channel allows it to interoperate with agents using other communication protocols. To verify this, we construct heterogeneous multi-agent groups mixing HyLaT- and NL-based agents in varying proportions and evaluate task performance and token usage. As shown in Figure 3(a), increasing the proportion of HyLaT agents consistently reduces token consumption while improving task performance, indicating that HyLaT agents can effectively communicate with NL agents through the shared text channel and guide the group toward more efficient and accurate outcomes. This compatibility is unavailable to pure latent communication methods, whose opaque signals cannot be interpreted by agents operating in natural language.

Robustness to Latent Perturbation To assess the stability, we inject Gaussian noise of varying magnitude into each latent vector at inference time. As shown in Figure 3(b), HyLaT maintains stable performance for $\sigma \leq 0.5$. This robustness stems from the hybrid channel design: while the latent channel carries intermediate reasoning, critical information such as the final answer is expressed through the text channel, where discrete tokens are more resilient to noise than continuous vectors, safeguarding the output from total degradation.

Heterogeneous Agent Communication HyLaT can be naturally extended to heterogeneous multi-agent settings with minimal architectural modification. During Stage 2 training, a lightweight MLP adapter is introduced for each agent to project received latent representations into its own semantic space prior to fusion, thereby bridging the representational gap between agents from different model families. To validate this, we construct a heterogeneous variant of HyLaT pairing Llama-3.2-1B-Instruct and Qwen2-1.5B-Instruct as communicating agents. We evaluate this variant on the MAD task under a 2-agent, 2-round setting. As

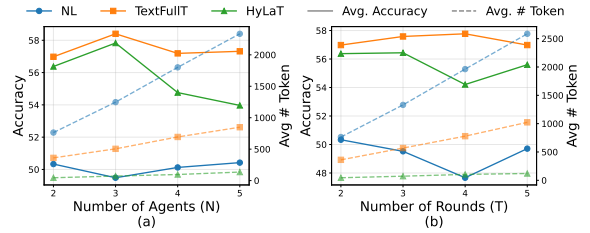


Figure 4: We vary the number of agents (N) while fixing the number of communication rounds to $T=2$ (left), and vary the number of rounds (T) while fixing $N=2$ (right).

shown in Figure 8(a), the results demonstrate that HyLaT successfully enables effective communication between agents from different model families, confirming the generality of our hybrid latent-text communication framework.

4.4 Scalability and Generalization Analysis

This section demonstrates that HyLaT can scale to more complex agent interactions and generalizes to other task scenarios, such as social simulation.

Scalability of Agent Count and Interaction Rounds

As shown in Figure 4, HyLaT’s token usage remains relatively stable as both N and T increase, while NL incurs rapidly growing overhead and TextFullT shows moderate growth. On accuracy, increasing N or T does not consistently lead to performance gains across all three methods, suggesting that effective coordination in more complex interactions generally remains challenging for 1B-scale models. For HyLaT, the degradation is also partially attributed to the gap between training and evaluation configuration: as the model is optimized for a small-scale setup where $N = 2$, $T = 2$, yet evaluated under larger-scale conditions. To verify this hypothesis, we constructed 3,400 training samples with 4-agent interactions and retrained Stage 2 of HyLaT. Figure 8(b) shows that the retrained model (HyLaT-scaled) notably alleviates the performance degradation. All these results demonstrate that **the efficiency advantage of HyLaT becomes increasingly pronounced as interactions scale, and that HyLaT can generalize to larger multi-agent settings, which can be further enhanced with correspondingly scaled training data.**

Generalization to Social Simulation To further explore the applicability of HyLaT to multi-agent social simulation, we conduct the *repeated trust game* from Xie et al. (2024), in which a trustor and a trustee interact over 7 rounds: the trustor decides how much to send each round, and the trustee decides how much to return. As shown in Table 3,

Method	Trust			
	Avg. Sent	Avg. Returned	# token	time
NL	3.82	5.10	2134.13	22.36
Autoform	4.56	7.75	4963.44	55.34
Ecolang	4.05	7.13	776.81	8.43
SDE	3.66	6.60	1717.69	18.82
TextFullT	3.22	9.04	797.94	8.45
HyLaT	3.53	10.84	182.00	3.49
Human*	7.48	12.24	-	-

Table 3: Results on *repeated trust game* for human trust behavior simulation. *Avg. Sent* and *Avg. Returned* represents the average amount sent by the trustors and that returned by the trustees. Human*: estimated from figures in (Cochard et al., 2004; Xie et al., 2024)

Question	Answer the question based on the given context. Context: Jordan gave people hugs whether he knew them or not. Question: How would Others feel as a result? Options: A. like kissing him; B. uncomfortable; C. happy Briefly explain your answer, and write your final choice on ...
Gen. Pos.	Top Attended Tokens
Round 1 First Latent	'< begin_of_text >', '<bot>', 'parentheses', 'answer', 'context', 'example', 'uncomfortable', 'enclosed'
Round 1 Text Answer	'< begin_of_text >', 'is', '<latent_0>', '<eot>', 'answer', 'parentheses', 'B', 'uncomfortable', '<bot>', '<latent_4>', '<latent_5>'
Round 2 First Latent	'< begin_of_text >', '<bot>', 'answer', '<latent_5>', '<latent_3>', '<latent_2>', '<latent_4>', '<latent_5>', '<latent_1>', 'parentheses', '<latent_4>', '<latent_1>', '<latent_2>', '<latent_5>'
Round 2 Text Answer	'< begin_of_text >', '<eot>', 'answer', 'parentheses', 'A', 'B', 'B', '<latent_4>', '<latent_5>', '<latent_3>', '<latent_2>', 'B', '<latent_1>'

Figure 5: Top attended words at various generation steps during multi-agent communication. Tokens are color-coded by source: black (instructions and questions), gray (templates), green/blue (agent’s own latent/text), and orange/red (other agents’ latent/text).

both HyLaT and TextFullT achieve substantially higher returned amounts than other methods. We attribute this to training on social reasoning data, which likely improves agents’ ability to understand and respond to others’ intentions. HyLaT additionally achieves this with far lower communication cost, suggesting that hybrid communication can preserve the social reasoning capability acquired while substantially reducing overhead.

4.5 Interpretability Analysis

Cross-Agent Attention Analysis To understand how agents utilize information from both channels during communication, we analyze the learned attention in Figure 5. In Round 1, latent generation primarily attends to keywords from the question and task instructions, while text generation shifts attention toward the agent’s own latent tokens. In Round 2, a more striking pattern emerges: latent generation is dominated by the other agent’s latent tokens from the previous round, providing direct evidence that cross-agent information change occurs

through these continuous representations. When generating the critical textual output, the agent attends to a mixture of the other agent’s textual response and its own latent tokens, indicating that the final decision integrates both explicitly communicated content and implicitly encoded details. Nevertheless, attention patterns alone do not represent causal explanations, and we will explore more rigorous interpretability analysis in future work.

Probing the Latent Tokens To examine whether latent tokens encode meaningful information, we visualize the final-step latent embeddings across five datasets using PCA. As shown in Figure 9, the latent representations exhibit clear domain-level clustering despite the model never receiving explicit domain labels, demonstrating that the latent channel organizes information by semantic content rather than surface form. The first two principal components account for 70.5% of the total variance, indicating that the latent space organizes around a small number of dominant semantic directions rather than distributing information uniformly. These observations suggest that latent tokens are not merely auxiliary padding: they actively encode task-specific semantics in a geometrically organized space, providing the downstream text decoder with structured, domain-relevant context.

5 Related Work

5.1 LLM-based Multi-agent Systems

LLM-based multi-agent systems (MAS) have demonstrated strong performance across various scenarios (Guo et al., 2024; Mou et al., 2024a). Representative applications include debate (Du et al., 2023; Tang et al., 2025), software development (Qian et al., 2024a), and social simulation (Mou et al., 2024b). In these settings, agents communicate via natural language, structured messages, or shared memory to exchange information and coordinate actions. However, as the number of agents and interaction rounds increases, the rapidly growing communication overhead poses efficiency challenges (Qian et al., 2024b; Chen et al., 2025).

5.2 Multi-agent Communication

To optimize the communication efficiency, early work on emergent communication (Lazaridou and Baroni, 2020) showed that agents can develop their own compact protocols to solve cooperative tasks. With the rise of LLMs, recent studies have investigated how to optimize communication for both

efficiency and performance. Some works (Chen et al., 2024; Mou et al., 2025; Chen et al., 2025) explore ways to compress or simplify message content while maintaining task effectiveness. Other works prune message-passing graph (Zhang et al., 2024) or reorganize protocols (Marro et al., 2024) to reduce redundancy. Despite these advances, most approaches are limited to either structured text or learned symbolic languages, which constrains the density of information that can be exchanged.

5.3 Latent Reasoning and Communication

Recent work has explored latent representations as an alternative medium for reasoning in LLMs (Hao et al., 2024; Shen et al., 2025; Wei et al., 2025), improving efficiency compared to explicit chain-of-thought approaches. While this line of research mainly focuses on single-turn reasoning, it has recently been extended to multi-agent collaboration. These methods typically enable semantic transfer across models by sharing hidden states or projecting KV-cache representations (Pham et al., 2023; Fu et al., 2025; Zou et al., 2025; Du et al., 2025). Despite these advances, existing approaches generally rely solely on latent communication, which reduces interpretability and controllability. Moreover, their thought-transfer mechanisms are often designed for sequential workflows and do not support multi-turn interactions, which are more common in general communication scenarios.

6 Conclusion

In this paper, we present HyLaT, a hybrid latent-text communication protocol that transfers elaborate cognitive signals through a latent channel and concise critical signals through a text channel. A two-stage training framework is proposed to enable agents to participate in effective multi-round hybrid communication. Experiments show that HyLaT dramatically improves communication efficiency while maintaining competitive task performance.

Limitations

While HyLaT demonstrates promising results, several limitations remain. We discuss them below and outline directions for future work.

- **Model scale.** Due to computational constraints, all experiments are conducted on relatively small models (1B and 3B parameters). Whether HyLaT’s benefits extend to larger-scale models remains to be verified.

- **Deployment scale.** HyLaT’s modified generation procedure is currently incompatible with inference frameworks such as vLLM, precluding evaluation in large-scale or long-horizon multi-agent scenarios, where communication efficiency becomes more important.
- **Text channel diversity.** The expressive diversity of the text channel is constrained by the scarcity of high-quality multi-agent communication data and human communication data. Currently, our training set mainly includes QA-style debating data, and following Shen et al. (2025), we append a fixed answer prompt to align generation, both of which limit the range of communicative behaviors the text channel can express. Exploring richer and more diverse communication scenarios is an important direction for future work.
- **Latent interpretability.** While HyLaT improves upon pure latent communication by preserving observable intermediate text outputs, the latent channel itself remains opaque. Decoding the semantic content of latent vectors is still highly challenging, and how to balance latent decodability with communication performance is a non-trivial open problem that we leave for future work.

References

- Weize Chen, Chenfei Yuan, Jiarui Yuan, Yusheng Su, Chen Qian, Cheng Yang, Ruobing Xie, Zhiyuan Liu, and Maosong Sun. 2024. Beyond natural language: LLMs leveraging alternative formats for enhanced reasoning and communication. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 10626–10641.
- Weize Chen, Jiarui Yuan, Chen Qian, Cheng Yang, Zhiyuan Liu, and Maosong Sun. 2025. Optima: Optimizing effectiveness and efficiency for LLM-based multi-agent system. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 11534–11557.
- Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. Think you have solved question answering? try arc, the ai2 reasoning challenge. *arXiv preprint arXiv:1803.05457*.
- Francois Cochar, Phu Nguyen Van, and Marc Willinger. 2004. Trusting behavior in a repeated investment game. *Journal of Economic Behavior & Organization*, 55(1):31–44.

- Yilun Du, Shuang Li, Antonio Torralba, Joshua B Tenenbaum, and Igor Mordatch. 2023. Improving factuality and reasoning in language models through multiagent debate. *arXiv preprint arXiv:2305.14325*.
- Zhuoyun Du, Runze Wang, Huiyu Bai, Zouying Cao, Xiaoyong Zhu, Yu Cheng, Bo Zheng, Wei Chen, and Haochao Ying. 2025. Enabling agents to communicate entirely in latent space. *arXiv preprint arXiv:2511.09149*.
- Tianyu Fu, Zihan Min, Hanling Zhang, Jichao Yan, Guohao Dai, Wanli Ouyang, and Yu Wang. 2025. Cache-to-cache: Direct semantic communication between large language models. *arXiv preprint arXiv:2510.03215*.
- Mor Geva, Daniel Khashabi, Elad Segal, Tushar Khot, Dan Roth, and Jonathan Berant. 2021. Did aristotle use a laptop? a question answering benchmark with implicit reasoning strategies. *Transactions of the Association for Computational Linguistics*, 9:346–361.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, and 1 others. 2024. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.
- Taicheng Guo, Xiuying Chen, Yaqi Wang, Ruidi Chang, Shichao Pei, Nitesh V Chawla, Olaf Wiest, and Xi-angliang Zhang. 2024. Large language model based multi-agents: A survey of progress and challenges. *arXiv preprint arXiv:2402.01680*.
- Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian. 2024. Training large language models to reason in a continuous latent space. *arXiv preprint arXiv:2412.06769*.
- Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. 2020. Constructing a multi-hop qa dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 6609–6625.
- Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Liang Wang, Weizhu Chen, and 1 others. 2022. Lora: Low-rank adaptation of large language models. *Iclr*, 1(2):3.
- Peter Jansen, Elizabeth Wainwright, Steven Marmorstein, and Clayton Morrison. 2018. Worldtree: A corpus of explanation graphs for elementary science questions supporting multi-hop inference. In *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)*.
- Di Jin, Eileen Pan, Nassim Oufattole, Wei-Hung Weng, Hanyi Fang, and Peter Szolovits. 2021. What disease does this patient have? a large-scale open domain question answering dataset from medical exams. *Applied Sciences*, 11(14):6421.
- Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William Cohen, and Xinghua Lu. 2019. Pubmedqa: A dataset for biomedical research question answering. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*, pages 2567–2577.
- Angeliki Lazaridou and Marco Baroni. 2020. Emergent multi-agent communication in the deep learning era. *arXiv preprint arXiv:2006.02419*.
- Samuele Marro, Emanuele La Malfa, Jesse Wright, Guohao Li, Nigel Shadbolt, Michael Wooldridge, and Philip Torr. 2024. A scalable communication protocol for networks of large language models. *arXiv preprint arXiv:2410.11905*.
- Peter R Monge and Noshir S Contractor. 2003. *Theories of communication networks*. Oxford University Press, USA.
- Xinyi Mou, Xuanwen Ding, Qi He, Liang Wang, Jingcong Liang, Xinnong Zhang, Libo Sun, Jiayu Lin, Jie Zhou, Xuanjing Huang, and 1 others. 2024a. From individual to society: A survey on social simulation driven by large language model-based agents. *arXiv preprint arXiv:2412.03563*.
- Xinyi Mou, Chen Qian, Wei Liu, Xuanjing Huang, and Zhongyu Wei. 2025. Ecolang: Efficient and effective agent communication language induction for social simulation. *arXiv preprint arXiv:2505.06904*.
- Xinyi Mou, Zhongyu Wei, and Xuan-Jing Huang. 2024b. Unveiling the truth and facilitating change: Towards agent-based large-scale social movement simulation. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 4789–4809.
- Joon Sung Park, Joseph O’Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. 2023. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th annual acm symposium on user interface software and technology*, pages 1–22.
- Chau Pham, Boyi Liu, Yingxiang Yang, Zhengyu Chen, Tianyi Liu, Jianbo Yuan, Bryan A Plummer, Zhaoran Wang, and Hongxia Yang. 2023. Let models speak ciphers: Multiagent debate through embeddings. *arXiv preprint arXiv:2310.06272*.
- Chen Qian, Wei Liu, Hongzhang Liu, Nuo Chen, Yufan Dang, Jiahao Li, Cheng Yang, Weize Chen, Yusheng Su, Xin Cong, and 1 others. 2024a. Chatdev: Communicative agents for software development. In *Proceedings of the 62nd annual meeting of the association for computational linguistics (volume 1: Long papers)*, pages 15174–15186.
- Chen Qian, Zihao Xie, Yifei Wang, Wei Liu, Kunlun Zhu, Hanchen Xia, Yufan Dang, Zhuoyun Du, Weize Chen, Cheng Yang, and 1 others. 2024b. Scaling large language model-based multi-agent collaboration. *arXiv preprint arXiv:2406.07155*.

- Maarten Sap, Hannah Rashkin, Derek Chen, Ronan Le Bras, and Yejin Choi. 2019. Social iqa: Commonsense reasoning about social interactions. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*, pages 4463–4473.
- Claude Elwood Shannon. 1948. A mathematical theory of communication. *The Bell system technical journal*, 27(3):379–423.
- Zhenyi Shen, Hanqi Yan, Linhai Zhang, Zhanghao Hu, Yali Du, and Yulan He. 2025. Codi: Compressing chain-of-thought into continuous space via self-distillation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 677–693.
- Aaditya Singh, Adam Fry, Adam Perelman, Adam Tart, Adi Ganesh, Ahmed El-Kishky, Aidan McLaughlin, Aiden Low, AJ Ostrow, Akhila Ananthram, and 1 others. 2025. Openai gpt-5 system card. *arXiv preprint arXiv:2601.03267*.
- Alon Talmor, Jonathan Herzig, Nicholas Lourie, and Jonathan Berant. 2019. Commonsenseqa: A question answering challenge targeting commonsense knowledge. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4149–4158.
- Yichen Tang, Weihang Su, Yujia Zhou, Yiqun Liu, Min Zhang, Shaoping Ma, and Qingyao Ai. 2025. Augmenting multi-agent communication with state delta trajectory. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 10230–10251.
- Xilin Wei, Xiaoran Liu, Yuhang Zang, Xiaoyi Dong, Yuhang Cao, Jiaqi Wang, Xipeng Qiu, and Dahua Lin. 2025. Sim-cot: Supervised implicit chain-of-thought. *arXiv preprint arXiv:2509.20317*.
- Shiguang Wu, Yaqing Wang, and Quanming Yao. 2025. Dense communication between language models. *arXiv preprint arXiv:2505.12741*.
- Chengxing Xie, Canyu Chen, Feiran Jia, Ziyu Ye, Shiyang Lai, Kai Shu, Jindong Gu, Adel Bibi, Ziniu Hu, David Jurgens, and 1 others. 2024. Can large language model agents simulate human trust behavior? *arXiv preprint arXiv:2402.04559*.
- Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D Manning. 2018. Hotpotqa: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 conference on empirical methods in natural language processing*, pages 2369–2380.
- Guibin Zhang, Yanwei Yue, Zhixun Li, Sukwon Yun, Guancheng Wan, Kun Wang, Dawei Cheng, Jeffrey Xu Yu, and Tianlong Chen. 2024. Cut the crap: An economical communication pipeline for llm-based multi-agent systems. *arXiv preprint arXiv:2410.02506*.
- Xinnong Zhang, Jiayu Lin, Xinyi Mou, Shiyue Yang, Xiawei Liu, Libo Sun, Hanjia Lyu, Yihang Yang, Weihong Qi, Yue Chen, and 1 others. 2025. Socioverse: A world model for social simulation powered by llm agents and a pool of 10 million real-world users. *arXiv preprint arXiv:2504.10157*.
- Jiaru Zou, Xiyuan Yang, Ruizhong Qiu, Gaotang Li, Katherine Tieu, Pan Lu, Ke Shen, Hanghang Tong, Yejin Choi, Jingrui He, and 1 others. 2025. Latent collaboration in multi-agent systems. *arXiv preprint arXiv:2511.20639*.

A Supplemented Implementation Details

A.1 Training Data Details

Datasets of Stage 1 To support hybrid communication, Stage 1 training requires data that naturally exhibits the two-part structure of HyLaT’s output: an elaborate cognitive signal encoding dense intermediate reasoning and a concise signal expressing the final response. Rather than constructing such data from scratch, we sample from training sets of existing datasets that already contain detailed reasoning processes alongside short final answers, treating the reasoning or explanation as the target for the latent channel and the concise answer as the supervision target for the text channel. Based on this, multiple QA pairs can be concatenated to form multi-turn dialogues. Dataset statistics are summarized in Table 4.

Datasets of Stage 2 To support multi-round multi-agent communication, we construct training data through multi-agent debate simulation along two complementary axes. (1) *Refinement*: In this setting, agents independently answer the same question in parallel and iteratively refine their responses through discussion. To ensure that inter-agent communication provides meaningful information gain, we deliberately avoid using a strong model for all rounds, as it tends to produce correct answers in the first round, leaving little room for correction through communication. Instead, we adopt a *weak-strong pairing strategy*: the first round is generated by Llama-3.2-1B and the second round by GPT-5, and dialogues in which the second round answer remains incorrect are discarded. We randomly sample questions from the Stage 1 datasets and retain 3,960 valid dialogues; (2) *Decomposition*: In this setting, agents collaboratively solve a complex question by first addressing complementary sub-problems in parallel, then synthesizing their findings to produce a final answer in the second round. We sample 2,000 structurally parallel questions from HotpotQA (Yang et al., 2018), 2WikiMultihopQA (Ho et al., 2020), and GSM8K-AUG-NL (Shen et al., 2025), respectively. The prompts for data generation are provided in Figure 6.

A.2 Evaluation Details

We implement all the multi-agent communication experiments using the framework provided by Tang et al. (2025). Following them, for datasets with

Dataset	Type	# samples
CommonsenseQA (Talmor et al., 2019)	commonsense reasoning	3,600
StrategyQA (Geva et al., 2021)	commonsense reasoning	1,800
SocialQA (Sap et al., 2019)	social reasoning	1,800
WorldTree (Jansen et al., 2018)	scientific question answering	2,204
PubMedQA (Jin et al., 2019)	medical question answering	500

Table 4: Statistics of training datasets in Stage 1.

Round 1: Initial Prompt

Please answer the following question: {question}

First explain your reasoning, and provide your final answer in the form `\boxed{answer}`, at the end of your response.

Round $t > 1$: Multi-Agent Interaction Prompt

These are the solutions from other agents:
One agent’s response: ````{agent_response}````
[repeated for each other agent]

Using the responses from other agents as additional information, can you provide your answer to the question? The original question is {question}.

Your final answer should be in the form `\boxed{answer}`, at the end of your response.

Figure 6: Prompts used for multi-agent interaction data construction. Round 1 elicits an initial response with explicit reasoning. In subsequent rounds, each agent receives the responses of all other agents and is asked to produce a revised answer.

more than 300 test examples, we evaluate on the first 300 samples. Evaluation tasks are constructed from the test splits of Stage 1 datasets as in-domain tasks, together with three additional out-of-domain datasets: MedQA (Jin et al., 2021), ARC-Easy and ARC-Challenge (Clark et al., 2018). Unless otherwise specified, all main experiments are conducted with $N = 3$ agents over $T = 2$ interaction rounds. During inference, we use the following generation configuration: temperature = 0.2, top- $k = 20$, top- $p = 0.9$, with sampling enabled. The prompt templates for multi-agent debate are adapted from Tang et al. (2025) (Figure 7).

A.3 Implementation Details

We follow previous work (Hao et al., 2024; Shen et al., 2025) to set the number of latent vectors $k = 6$, and we list the hyper-parameter settings used in Table 7. For Stage 2 training, we only supervise on the last turn, considering that the intermediate conclusions can be incorrect in the refinement data.

Method	In-Domain										Out-of-Domain						Inference Efficiency	
	Commonsense		StrategyQA		SocialIQA		WorldTree		PubMedQA		MedQA		ARC-E		ARC-C			
	Avg.	Maj.	Avg.	Maj.	Avg.	Maj.	Avg.	Maj.	Avg.	Maj.	Avg.	Maj.	Avg.	Maj.	Avg.	Maj.	# token	time
Training-Free Single-Agent Baseline																		
Single-Agent	73.67	73.67	66.00	66.00	74.00	74.00	89.00	89.00	71.00	71.00	62.00	62.00	42.67	42.67	45.67	45.67	130.65	2.46
Training-Free Text-Based Communication Methods																		
NL	72.33	76.00	66.56	67.67	74.89	76.33	89.78	90.33	69.00	69.00	60.44	62.00	44.00	44.67	44.89	46.00	1094.83	19.13
AutoForm	68.11	69.00	70.33	72.67	75.56	76.00	88.78	90.00	69.56	69.33	65.00	68.00	42.56	43.00	44.22	44.33	1668.19	29.09
EcoLang	72.33	74.33	68.44	68.00	74.44	73.33	87.89	88.33	69.22	69.33	61.11	61.33	43.00	43.67	42.78	42.00	603.64	10.43
SDE	72.56	74.00	64.89	65.33	72.89	74.67	86.56	88.67	69.00	69.00	58.67	62.67	42.56	43.67	43.67	44.67	913.33	16.35
Training-Free Latent-Space Communication Methods																		
Cipher	71.67	73.67	66.89	69.33	74.89	75.00	86.56	87.00	67.89	69.33	59.56	60.33	44.11	44.33	46.11	48.00	1080.19	18.40
LatentMAS-V	70.78	72.17	59.67	61.67	73.00	74.33	84.56	84.67	62.44	64.00	60.11	61.67	41.78	41.33	42.89	44.33	570.63	10.58
Training-Based Communication Methods																		
TextFullIT	72.22	72.67	77.22	78.67	84.78	84.33	87.67	89.33	74.00	75.33	60.44	62.00	43.22	43.33	44.89	46.00	748.74	12.79
LatentFullIT	78.67	78.33	81.22	81.00	89.22	89.67	84.67	84.67	73.56	73.33	59.44	59.00	43.67	43.67	43.33	43.33	57.00	1.21
HyLaT	78.67	78.33	85.00	85.00	90.11	90.00	86.89	86.33	73.89	74.00	59.44	59.33	44.00	44.00	43.44	43.67	72.00	2.46

Table 5: Experimental results of different methods on Llama-3.2-3B-Instruct. We report average accuracy (Avg.) and majority-vote accuracy (Maj.) across agents, along with communication efficiency measured by average token count (# token) and wall-clock time (time, in seconds). **Bold** denotes the best result and underline denotes the second best.

Method	In-Domain										Out-of-Domain						Inference Efficiency	
	Commonsense		StrategyQA		SocialIQA		WorldTree		PubMedQA		MedQA		ARC-E		ARC-C		# token	time
	Avg.	Maj.	Avg.	Maj.	Avg.	Maj.	Avg.	Maj.	Avg.	Maj.	Avg.	Maj.	Avg.	Maj.	Avg.	Maj.		
Training-Free Single-Agent Baseline																		
Single-Agent	58.00	58.00	50.67	50.67	68.67	68.67	72.00	72.00	61.33	61.33	39.00	39.00	38.00	38.00	35.00	35.00	87.11	<u>1.65</u>
Training-Free Text-Based Communication Methods																		
NL	59.33	60.67	50.33	50.00	67.22	69.00	74.33	77.00	58.11	58.00	37.78	38.33	37.67	38.33	38.00	39.67	552.40	10.38
AutoForm	50.78	55.33	50.22	50.67	66.56	68.00	77.89	79.67	56.78	57.33	35.33	36.33	37.67	38.33	35.33	36.67	344.69	6.51
EcoLang	61.11	62.00	51.00	51.00	69.22	69.33	78.56	78.67	59.00	58.67	36.22	36.67	36.67	37.33	39.67	39.67	235.55	4.49
SDE	58.67	61.33	49.67	49.33	67.56	70.33	75.44	78.33	59.44	59.67	<u>38.78</u>	40.67	38.00	39.33	37.22	38.33	1514.63	10.29
Training-Free Latent-Space Communication Methods																		
Cipher	60.11	60.00	51.67	51.67	69.78	70.00	78.33	78.67	<u>61.11</u>	61.00	36.00	36.33	39.33	39.33	36.11	36.33	691.38	7.84
LatentMAS-V	4.56	4.00	0.33	0.00	13.78	13.00	10.33	10.00	<u>21.00</u>	20.33	13.89	12.33	4.33	3.67	4.33	3.00	587.99	11.86
Training-Based Communication Methods																		
TextFullIT	<u>62.56</u>	<u>62.00</u>	55.11	55.00	76.11	76.00	78.33	78.67	59.11	60.33	38.67	<u>39.00</u>	38.56	40.00	37.67	39.33	545.30	10.06
LatentFullIT	59.83	60.00	<u>55.17</u>	<u>56.67</u>	<u>76.17</u>	<u>76.33</u>	76.00	75.33	53.67	53.33	36.50	38.33	<u>41.00</u>	<u>41.00</u>	37.67	38.67	46.00	0.81
HyLaT	62.78	63.00	55.89	56.67	76.44	76.33	78.56	<u>78.67</u>	59.67	<u>61.00</u>	38.67	38.67	41.89	41.67	<u>39.44</u>	39.67	<u>66.00</u>	2.43

Table 6: Experimental results of different methods on Qwen2-1.5B-Instruct. We report average accuracy (Avg.) and majority-vote accuracy (Maj.) across agents, along with communication efficiency measured by average token count (# token) and wall-clock time (time, in seconds). **Bold** denotes the best result and underline denotes the second best.

Configuration	Stage 1	Stage 2
Model initialization	Llama-3.2-1B-Instruct	Stage 1
LoRA	rank=128, alpha=32	rank=128, alpha=32
Global batch size	128	128
Peaking learning rate	8.00e-04	3.00e-04
Optimizer	AdamW	AdamW
LR scheduler	Cosine	Cosine
Training epochs	3	3
Warmup ratio	0.03	0.03
Precision	bfloat16	bfloat16
α	1	1
β	1	1
γ	20	20

Table 7: The detailed training hyper-parameters of HyLaT.

B Supplementary Experimental Results and Analysis

B.1 Heterogeneous Agent Communication

We implemented a heterogeneous variant of HyLaT pairing Llama-3.2-1B and Qwen2-1.5B as communicating agents, with a lightweight MLP adapter bridging their respective latent spaces. We evaluated this variant on the MAD task under a 2-agent and 2-round setting. Figure 8(a) shows that although the heterogeneous setting sometimes does not surpass the best homogeneous configuration (2 $\times$ stronger model), HyLaT consistently performs well across all agent configurations.

B.2 Further Improve the Scalability

In Figure 4(a), we can observe that performance begins to degrade when number of agents exceeds 3. This degradation with more agents is attributed to

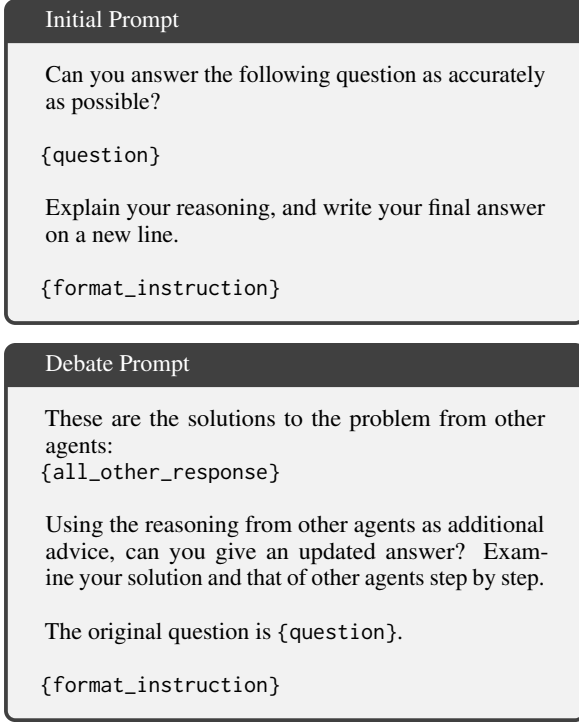


Figure 7: Prompt templates used for multi-agent debate evaluation.

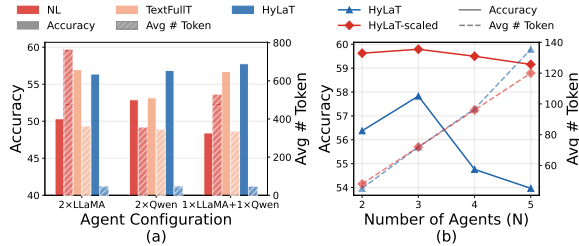


Figure 8: (a) Performance of different communication methods under homogeneous and heterogeneous agent configurations; (b) Performance of HyLaT trained on original vs. scaled data as the number of agents increases.

two factors: (1) this phenomenon is a general characteristic of multi-agent systems at this model scale or for these tasks, rather than a limitation introduced by our method alone, as TextFullT exhibits the same degradation pattern when the number of agents exceeds 3. (2) the degradation in HyLaT is partly due to a gap between training and evaluation: Our current training setup uses 2-agent interaction, which does not fully cover the distribution of larger-scale agent configurations at evaluation time. To verify this hypothesis and mitigate the issue, we constructed 3,400 training samples with 4-agent interactions using the *refinement* method in Sec 3.1, and retrained stage 2 of HyLaT. Results in Figure 8(b) show that such training data notably alleviates the performance degradation, confirming

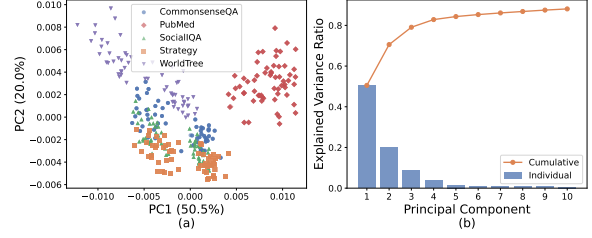


Figure 9: (a) PCA analysis of final-step latent embeddings across five datasets. (b) Explained variance ratio of the top-10 principal components of the latent embeddings.

that the scalability limitation stems from insufficient training coverage rather than a fundamental constraint of our method.

B.3 Probing the Latent Tokens

To probe the semantic content encoded in HyLaT’s latent channel, we collect 180 latent embeddings produced by agents at the final interaction round for each for the five datasets and apply PCA to project them into two dimensions. The results are visualized in Figure 9. As shown in the figure, embeddings from the same dataset form relatively compact clusters that are well-separated from those of other datasets, suggesting that the latent vectors capture dataset-specific semantic structure rather than collapsing into an undifferentiated representation. This provides indirect evidence that the latent channel encodes meaningful content that reflects the nature of the task and communicative context, even though the vectors are never decoded into discrete tokens during inference.

B.4 Different Model Scales

We extend the experiments on larger Llama backbones, i.e., Llama-3.2-3B-Instruct. Results in Table 5 show that HyLaT scales effectively to this setting: task performance remains competitive across most benchmarks, and the efficiency advantage is fully preserved. The exception is WorldTree, where the 3B model already achieves strong performance under text-based communication, leaving limited room for further improvement.

B.5 Different Model Families

We also conducted additional experiments using Qwen2-1.5B-Instruct as the backbone. Results in Table 6 show that HyLaT also demonstrates effectiveness on Qwen, matching or outperforming TextFullT across most tasks, while achieving dramatically higher communication efficiency.

We also note that LatentMAS-V fails to produce meaningful results on most MAD tasks on Qwen, suggesting that adapting the original approach to multi-round multi-agent communication for different models requires non-trivial effort. Overall, the aforementioned results demonstrate that HyLaT possesses superior cross-family generalization.